Networking Assignment - Subnetting Solutions

192.168.10.0/24 Subnet Division and Device Allocation

Question 1: Divide 192.168.10.0/24 into 4 Equal Subnets

Question: Given a network address of 192.168.10.0/24, divide it into 4 equal subnets and write down the subnet mask, network address, and broadcast address for each subnet.

Answer:

Given network: 192.168.10.0/24

To create 4 equal subnets, we borrow 2 bits because $2^2 = 4$.

- New CIDR: /26
- Subnet mask: 255.255.255.192
- Total addresses per subnet: $2^6 = 64$
- Usable hosts per subnet: 64 - 2 = 62

Subnet	Subnet Mask	Network Address	Broadcast Address
1	255.255.255.192	192.168.10.0	192.168.10.63
2	255.255.255.192	192.168.10.64	192.168.10.127
3	255.255.255.192	192.168.10.128	192.168.10.191
4	255.255.255.192	192.168.10.192	192.168.10.255

Each subnet has 62 usable host IP addresses.

Question 2: Assign IP Addresses to 5 Devices

Question: For one of the subnets you created, assign valid IP addresses to 5 devices (such as a router, two laptops, a printer, and a mobile) and list which IP goes to which device. Hint: Make sure not to use the network or broadcast address for any device.

Answer:

I selected Subnet 1 (192.168.10.0/26) to connect five devices.

- Network address: 192.168.10.0
- Subnet mask: 255.255.255.192
- Default gateway: 192.168.10.1
- Usable host range: 192.168.10.1 – 192.168.10.62
- Broadcast address: 192.168.10.63

Device	IP Address	Subnet Mask	Default Gateway
Router	192.168.10.1	255.255.255.192	Not applicable
Laptop 1	192.168.10.10	255.255.255.192	192.168.10.1
Laptop 2	192.168.10.11	255.255.255.192	192.168.10.1
Printer	192.168.10.12	255.255.255.192	192.168.10.1
Mobile	192.168.10.13	255.255.255.192	192.168.10.1

Conclusion: All five devices have unique IP addresses in the same subnet. None uses the network or broadcast address.

Question 3: Draw a Subnet Plan Diagram

Question: Draw a simple diagram (hand-drawn or digital) showing your subnet plan: label each subnet, the range of IPs, and where each device is connected.

Answer:

Subnet Plan Structure / Diagram Representation:

Component / Network Level	**Details & Assigned IPs**
Internet Connection	Connected via Gateway Router
Router (Gateway)	192.168.10.1
Active Subnet 1	192.168.10.0/26 (Usable IPs: 192.168.10.1 – 192.168.10.62)
Connected Devices in Subnet 1	• Laptop 1: 192.168.10.10 • Laptop 2: 192.168.10.11 • Printer: 192.168.10.12 • Mobile: 192.168.10.13
Other Available Subnets	• Subnet 2: 192.168.10.64/26 • Subnet 3: 192.168.10.128/26 • Subnet 4: 192.168.10.192/26
Broadcast Address	192.168.10.63 (Subnet 1)

Note: You can draw this exact topology diagram in your notebook or graphics software based on the table above.

Question 4: Explain the Subnetting Calculation Step by Step

Question: Document step-by-step how you calculated the subnet mask, number of hosts per subnet, and assigned IP addresses for your plan, as if you are explaining it to a friend who is new to subnetting.

Answer:

Step 1: Find the number of required subnets.

We need 4 subnets. Therefore, we borrow 2 bits from the host portion because $2^2 = 4$.

Step 2: Calculate the new subnet mask.

The original network is /24. We add the 2 borrowed bits:
24 + 2 = 26
The new subnet mask is 255.255.255.192.

Step 3: Calculate the number of hosts.

The number of host bits remaining is:
32 - 26 = 6
Total addresses per subnet: $2^6 = 64$
Usable host addresses: 64 - 2 = 62
(We subtract 2 because one address is reserved for the network address and one for the broadcast address).

Step 4: Find the subnet increment.

The last subnet-mask octet is 192.
256 - 192 = 64
Therefore, the subnet network addresses increase by 64: .0, .64, .128, and .192.

Step 5: Find the broadcast addresses.

The broadcast address is one less than the next subnet's network address. So the broadcast addresses are .63, .127, .191, and .255.

Step 6: Assign device IPs.

Choose usable addresses from Subnet 1 and assign them to the router, two laptops, printer, and mobile. Set the router's address as the default gateway for the other devices.

Conclusion: The 192.168.10.0/24 network is divided into four equal /26 subnets, each supporting 62 usable host addresses.